Journal of Imaging Informatics in Medicine (2025) 38:3110–3124
https://doi.org/10.1007/s10278-025-01416-7

ORIGINAL PAPER

Multi-class Classification of Retinal Eye Diseases from Ophthalmoscopy Images Using Transfer Learning-Based Vision Transformers

Elif Setenay Cutur[1] · **Neslihan Gokmen Inan**[2]

Received: 11 September 2024 / Revised: 12 January 2025 / Accepted: 14 January 2025 / Published online: 27 January 2025
© The Author(s) under exclusive licence to Society for Imaging Informatics in Medicine 2025

Abstract

This study explores a transfer learning approach with vision transformers (ViTs) and convolutional neural networks (CNNs) for classifying retinal diseases, specifically diabetic retinopathy, glaucoma, and cataracts, from ophthalmoscopy images. Using a balanced subset of 4217 images and ophthalmology-specific pretrained ViT backbones, this method demonstrates significant improvements in classification accuracy, offering potential for broader applications in medical imaging. Glaucoma, diabetic retinopathy, and cataracts are common eye diseases that can cause vision loss if not treated. These diseases must be identified in the early stages to prevent eye damage progression. This paper focuses on the accurate identification and analysis of disparate eye diseases, including glaucoma, diabetic retinopathy, and cataracts, using ophthalmoscopy images. Deep learning (DL) has been widely used in image recognition for the early detection and treatment of eye diseases. In this study, ResNet50, DenseNet121, Inception-ResNetV2, and six variations of ViT are employed, and their performance in diagnosing diseases such as glaucoma, cataracts, and diabetic retinopathy is evaluated. In particular, the article uses the vision transformer model as an automated method to diagnose retinal eye diseases, highlighting the accuracy of pre-trained deep transfer learning (DTL) structures. The updated ViT#5 model with the augmented-regularized pre-trained model (AugReg ViT-L/16_224) and learning rate of 0.00002 outperforms the state-of-the-art techniques, obtaining a data-based accuracy score of 98.1% on a publicly accessible retinal ophthalmoscopy image dataset, which includes 4217 images. In most categories, the model outperforms other convolutional-based and ViT models in terms of accuracy, precision, recall, and F1 score. This research contributes significantly to medical image analysis, demonstrating the potential of AI in enhancing the precision of eye disease diagnoses and advocating for the integration of artificial intelligence in medical diagnostics.

Keywords Multi-class classification · Deep learning · Convolutional neural networks · Vision Transformer · Retinal images

Introduction

Eyes are very delicate organs of the human body, and their level of wellness affects one's daily life. Therefore, early detection and treatment of any eye disease is crucial. There are numerous ways to identify any kind of eye disease, but one of the most popular ones is to look at the ophthalmoscopy (fundoscopy) images. This technique looks into the back of the eye to check the retina. This examination helps diagnose conditions and identify risk factors for potential vision loss associated with the retina [1].

Machine learning (ML) and artificial intelligence (AI) models are integrating into our lives day by day with the development of technology. Using AI in medicine is spreading more and more, from detecting diseases in the patient's body to assisting doctors in surgical operations. AI can also be used to examine retinal images to diagnose eye diseases, even in their early stages [2, 3]. This study aims to use deep learning (DL) and image recognition models to classify retinal images obtained from ophthalmoscopy examinations according to the relative disease type of glaucoma, cataract, diabetic retinopathy, and normal patients. In the following section, glaucoma, cataracts, and diabetic retinopathy examined in this study are briefly explained to provide some insight into the importance of detecting these diseases.

✉ Neslihan Gokmen Inan
ninan@ku.edu.tr

[1] Graduate School of Sciences and Engineering, Data Science, Koç University, Istanbul, Turkey

[2] College of Engineering, Department of Computer Engineering, Koç University, Rumelifeneri Yolu, 34450 Sarıyer, Istanbul, Turkey

Glaucoma is an ocular condition that has the potential to lead to vision impairment and eventual blindness by causing harm to the optic nerve situated at the rear of the eye. The indications of glaucoma may manifest gradually, making them imperceptible to the patient. The sole means of detecting glaucoma is through a comprehensive dilated eye examination. While there is no cure for this condition, early intervention can frequently halt the progression of damage and preserve one's eyesight [4].

Cataract refers to a cloudy region within the eye's lens, the transparent part responsible for focusing light. Initially, individuals may not be aware of the presence of a cataract. However, as time passes, cataracts can cause vision to become blurry, obscured, or less vibrant. This can result in difficulties with activities such as reading and everyday tasks. Gradually, cataracts may lead to a decline in vision. Fortunately, cataracts can be treated through surgical operation [4].

Diabetic retinopathy is an eye disorder that can result in vision impairment and even blindness among individuals with diabetes. This condition specifically impacts the blood vessels within the retina, which is the light-sensitive layer of tissue situated at the back of the eye.

In its initial stages, diabetic retinopathy may not show any noticeable symptoms. However, early detection is crucial as it allows patients to take proactive measures to protect their vision [4].

Achieving precise classification of eye diseases necessitates a high level of clinical knowledge and expertise [1]. Recently, various AI-based decision support systems have been developed to mitigate laborious workloads and reduce misdiagnosis rates [2, 3, 5]. However, further research is crucial to the development and validation of these algorithms, integrating clinical standards alongside advancements in DL methodologies. In this study, our contribution can be summarized as follows:

1. We propose a study that shows the importance of transfer learning-based evaluation in classifying the ophthalmoscopy images.
2. We compare different CNN models and VIT models both with each other and within themselves.
3. We investigate several approaches that can achieve notable performance while minimizing computational costs and limited runtime choices availability to lower execution costs and high GPU demand.

In this context our study is organized as follows. The "Literature Survey" section presents the literature review. The "Materials and Methods" section describes the dataset, utilized DL architectures, and evaluation metrics, and in the "Results and Discussion" section, the details of the experimental results and discussion about convolutional neural network (CNN) and vision transformer (ViT) implementation and transfer learning algorithms are presented. Finally, the paper is concluded by summarizing the key points in the "Conclusion" section.

Literature Survey

In ophthalmoscopy imaging, the disease recognition procedure is crucial for early detection and treatment. Literature available for eye disease classification can be divided into two categories. The first category deploys traditional image processing techniques and ML. The second category uses various DL methods. The discussion of the recent studies is summarized in Table 1.

Zeng et al. (2019) proposed a Siamese-like CNN model based on the Inception V3 architecture for diabetic retinopathy detection, achieving a Kappa score of 0.829 on the EyePACS dataset comprising 35,126 images. Their results highlighted the utility of tailored CNN architectures for retinal disease detection [6]. Pan et al. (2020) extended this approach by using DenseNet, ResNet50, and VGG16 models to classify four diabetic retinopathy lesions, achieving AUC scores as high as 0.96 for specific categories. This study demonstrated the potential of CNNs for lesion-specific classification using fundus fluorescein angiography images [7]. Building on these efforts, Gour and Khanna (2021) utilized transfer learning-based CNN models for multi-class, multi-label classification of fundus images from the ODIR database, which included eight disease categories. Among the models, VGG16 paired with the SGD optimizer showed the best performance, emphasizing the importance of pretrained architectures and optimizer choices in ophthalmological disease detection [8]. Badah et al. (2022) expanded the scope by integrating traditional machine learning classifiers with CNNs for glaucoma detection, finding that a ResNet152-based CNN significantly outperformed traditional methods with an accuracy of 84% [9].

Recent advancements have focused on enhancing model performance and robustness. Karthik et al. (2023) introduced a novel CNN approach for optical coherence tomography (OCT) image classification, which emphasized high-contrast retinal regions and achieved improved accuracy on imbalanced datasets. Their method demonstrated the potential of CNN enhancements for addressing class imbalance [10]. Similarly, Dutta et al. (2023) developed a hybrid model, Conv-ViT, that combined CNNs and vision transformers to leverage both texture and shape-based feature extraction. Their model achieved a weighted average accuracy of 94% for retinal disease detection, illustrating the benefits of integrating complementary DL architectures [11]. Nguyen et al. (2024) proposed a ResNet152-based system for processing ultra-wide-field fundus imaging (UFI) data, achieving

Table 1 Summary of recent advances in deep learning-based models

Study	Dataset	Method	Results	Limitation	Year	Country
Zeng et al. [6]	EyePACS Diabetic Retinopathy	Siamese-like CNN model	AUC: 0.951 Kappa: 0.829	Binocular model difficulties	2019	China
Pan et al. [7]	Eye Center of Zhejiang University School of Medicine	DenseNet, ResNet50, VGG16	AUC: 0.96 (for laser scars)	Data may not be generalizable (local hospital)	2020	China
Gour and Khanna [8]	Ocular Disease Intelligent Recognition (ODIR)	Transfer learning-based CNN	AUC: 0.689	Class-imbalance problems	2021	India
Badah et al. [9]	ODIR	ML and CNN models	Accuracy: 0.840	Binary classification for glaucoma	2022	Saudi Arabia
Karthik et al. [10]	OCT-C8 dataset	ResNet50	Accuracy: 0.875	Specific dataset characteristics	2023	India
Dutta et al. [11]	Optical coherence tomography image database (OCTID)	Conv-ViT	Accuracy: 0.940	Complexity and need for high GPU	2023	Bangladesh
Nguyen et al. [12]	4697 Ultra-wide-field fundus imaging (UFI)	Self-Supervised Learning ResNet152	AUC: 0.965	Specific dataset characteristics	2024	Republic of Korea
Benbakretia et al. [13]	Kaggle Dataset—4 classes (fundus images)	GoogleNet	Accuracy: 0.927	Smoother loss function	2024	Algeria

an AUC of 96.47%. This approach also demonstrated the value of feature enhancement and visualization techniques for assisting ophthalmologists in clinical diagnosis [12]. Benbakretia et al. (2024) leveraged pretrained CNN models, including AlexNet and GoogleNet, to classify diabetic retinopathy, cataracts, and glaucoma from fundus images, achieving a top accuracy of 92.7% with GoogleNet. This study underscored the importance of data augmentation and optimizer selection in enhancing DL model performance for ophthalmic applications [13].

Materials and Methods

Dataset

The publicly available eye disease dataset used for this study is collected from the Kaggle [14]. The dataset consists of retinal ophthalmoscopy images belonging to one of four different categories: normal (N), glaucoma (G), cataract (C), and diabetic retinopathy (DR). The retinal ophthalmoscopy images are obtained from different resources, such as HRF, IDRiD, and Oculus recognition. Despite the availability of a Kaggle dataset containing 35,136 images, we opted to use a curated subset of 4217 images. This decision was based on balancing data diversity with computational feasibility, as processing the entire dataset would have exceeded available hardware capabilities. The selected subset maintains representational diversity across disease categories, ensuring effective model training while remaining computationally manageable given the study's resource constraints. Therefore, the normal category consists of 1074 images, the glaucoma category contains 1007 images, the cataract category contains 1038 images, and lastly, diabetic retinopathy consists of 1098 images. Table 2 displays the distribution of the various classes in order to explore any class imbalances, which may have an impact on model performance. The normal class takes 25.5% of the whole dataset, while glaucoma takes 26%, cataract takes 23.9%, and diabetic retinopathy takes 24.6%. A subset of the Kaggle dataset was selected to balance the data and reduce computational costs. Using the entire dataset would significantly increase training and inference times, even on advanced hardware such as the A100 GPU, which supports approximately 2500 inferences per day. The balanced subset was chosen to represent the original dataset's class distribution and ensure that the evaluation metrics remain robust. Although high precision and F1 scores may result from balancing, random selection ensures representativeness, mitigating potential bias.

Table 2 Distribution of retinal eye diseases

Classes	*N*	%
Normal	1074	25.47
Glaucoma	1007	23.88
Cataracts	1038	24.61
Diabetic retinopathy	1098	26.04

Random snippets from the images in this dataset can be seen in Fig. 1. All images are obtained in three channels (RGB).

The original dataset includes different sizes of images, which were resized in the preprocessing phase into 224×224 per 4217 images within three channels. This huge number of pixels overwhelms the GPU memory and creates limitations when designing the network architecture and selecting the hyperparameters. To overcome this situation, the total 4217×224×224×3 volume is divided into 24 small patches to minimize memory consumption.

DL Architectures

Four CNN and six ViT architectures are used to achieve high performance in multi-class retinal eye disease classification. The base CNN model consists of three convolutional layers, with a pooling layer for each. In addition to that, it consists of one flattening layer, two dense layers, and one dropout layer. It also has one rescaling layer, which makes a total number of ten for the layers of the base CNN model.

The CNN architectures used are ResNet50 [15], DenseNet121 [16], and Inception-ResNetV2 [17], known for their robust feature extraction capabilities. Inception-ResNet-v2 architecture merges the Inception module—with its parallel convolutions of varying filter sizes to grasp features at multiple scales—with residual connections, simplifying the training of exceptionally deep networks. Figure 2 illustrates Inception-ResNet-v2 which has an enhanced depth architecture. With more than 560 layers, Inception-ResNet-v2 delivered unparalleled performance across various benchmark datasets upon its debut [18].

Vision transformers (ViTs) exploit both self-attention and cross-attention mechanisms, allowing them to capture dependencies over both short and long distances within the image. This dual capability enables ViTs to extract nuanced spatial relationships, making them particularly effective for complex image classification tasks, as evidenced in recent studies where both local and global feature extraction improved segmentation accuracy [19].

The ViT model architecture was proposed by Dosovitskiy et al. in a groundbreaking paper released in 2020 [20]. This approach allows the ViT to capture global dependencies between any parts of the image directly, leveraging the self-attention mechanism of transformers. The original paper introduces several configurations of ViT, among

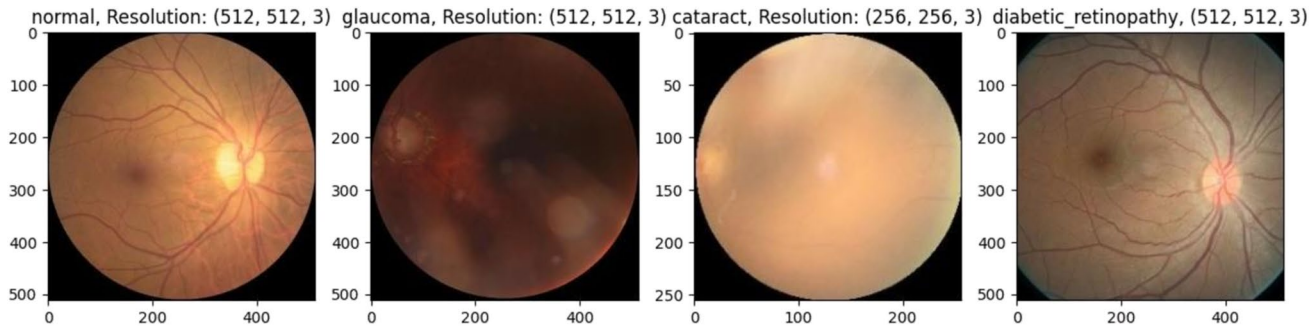


Fig. 1 Sample ophthalmoscopy images of classes

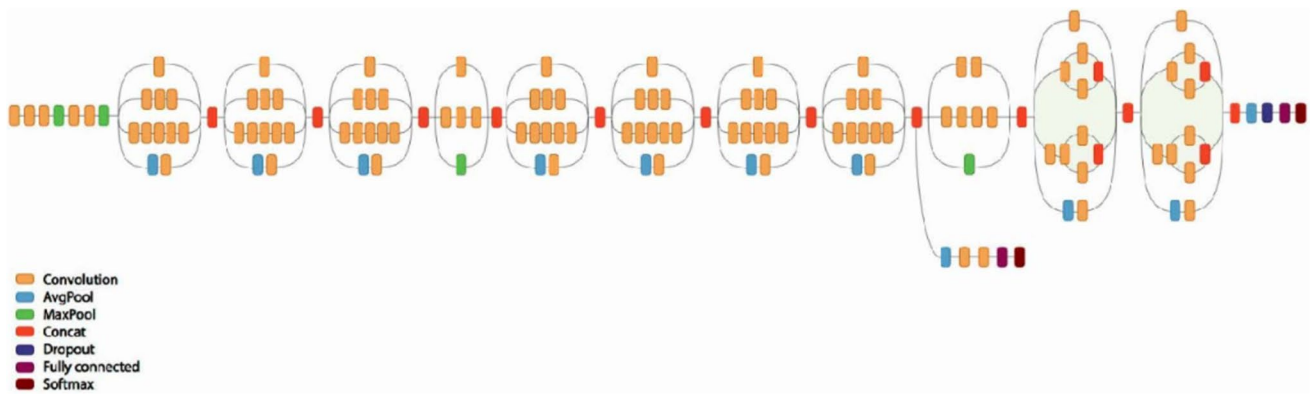


Fig. 2 Architecture of Inception-ResNet-v2 Network [18]

which the ViT-Base-16 (ViT-B/16), ViT-Large-16 (ViT-L/16), and a variant of ViT-L/16 specifically designed for smaller resolution inputs, ViT-L/16 for 224 × 224 pixels (ViT-L/16–224), are notable [20]. These three pre-trained models are used and compared in this study. ViT-Base-16 (ViT-B/16) configuration is considered the base model. It is designed with a moderate number of parameters and is a good starting point for understanding the balance between performance and complexity. The "16" in ViT-B/16 refers to the size of the patches each image is divided into, with each patch being 16 × 16 pixels. ViT-Large-16 (ViT-L/16) is a larger and more complex version of the ViT model, designed for higher performance at the cost of increased computational requirements. Like the base model, it also processes the image in 16 × 16 pixel patches but includes more layers and a larger number of attention heads in the transformer encoder, providing it with a greater capacity to learn from data. ViT-Large-16 for 224 × 224 pixels (ViT-L/16–224) variant is specifically optimized for processing images of 224 × 224 pixels resolution. It maintains the large scale and complexity of the ViT-L/16 configuration but is fine-tuned for performance on smaller resolution inputs, a common standard in image recognition tasks. The innovative approach of using transformers for image processing by the ViT model demonstrates the versatility and potential of transformer architectures beyond NLP, offering a fresh perspective on handling visual data and setting new benchmarks for image classification tasks [20]. Figure 3 demonstrates the proposed ViT architecture for multi-class retinal eye disease classification from ophthalmoscopy images.

Training and Inference

The training and test data used in this project has a batch size of 24. The training and testing of the CNN models were conducted using the Keras module from TensorFlow. The ViT models were trained and tested using Torch module, since it contains a variety of pre-trained ViT models. In this study, we utilized various models pre-trained on ImageNet, a general-purpose image dataset. The pre-trained weights provide general visual feature representations, which we fine-tuned on our specific dataset of retinal images. No separate domain-specific pre-training on an ophthalmoscopy dataset was conducted; instead, the ImageNet-pre-trained weights were directly adapted to our ophthalmology dataset through fine-tuning. This approach leverages the general feature extraction capabilities of the ViT model, transferring them effectively to the retinal classification task. The pre-trained models have 16 patches, which indicates the input images are divided into patches of 16 × 16 pixels. The vision transformer (ViT) models used in this study were pre-trained on the ImageNet dataset, a large-scale general-purpose image dataset. This pre-training provides robust initial weights, which were fine-tuned on our retinal disease classification dataset. For VIT#5 and VIT#6, augmented-regularized pre-trained model (ViT-L/16_224) is used, which additionally expects input images of 224 × 224 pixels in resolution. Five of the ViT models used the Adam optimizer, which is a popular optimizer in image recognition projects. Only ViT#6 had the RAdam optimizer to observe the impact of using Adam as an optimizer. For the learning rate, different values of 0.0001, 0.00002, and 0.000001 were used to measure the importance of learning rate under the same given conditions.

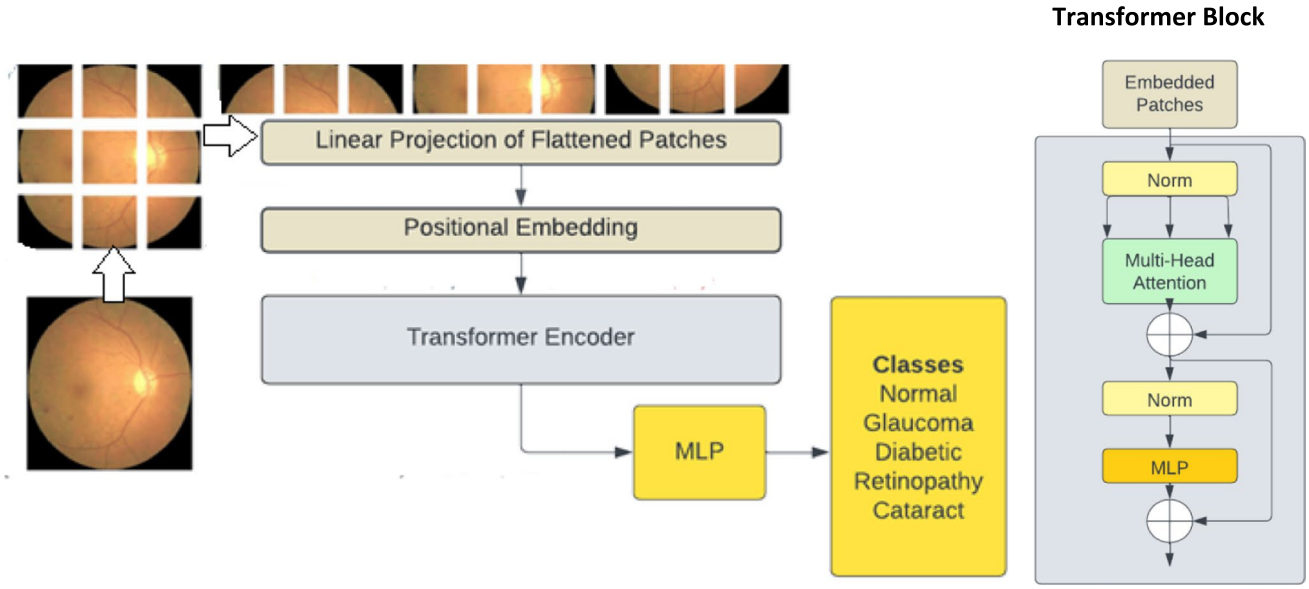


Fig. 3 Architecture of proposed ViT network

Configurations that differ each ViT model from each other can be observed in Table 3. Since ViT models require more time and source to train, High-RAM option with a system RAM value of 51 GB was used as the runtime settings. Experiments were conducted using an NVIDIA A100 GPU, chosen for its high processing power to handle vision transformer (ViT) and CNN models. The inference time for ViTs averaged approximately 34 s per image and 52 min overall, while the CNN models achieved faster results, averaging around 37 min overall. This notable difference in inference speed emphasizes the computational demand of ViTs compared to CNNs. Loss functions significantly impact model performance in classification tasks, particularly in medical image analysis. In this study, we opted for the cross-entropy loss function, which is commonly used and computationally efficient for multi-class classification. Although various hybrid loss functions, such as those combining focal Tversky, binary cross-entropy, and Jaccard index, have recently been used to improve segmentation accuracy by balancing class representation and enhancing regional consistency through refined pixel- and region-level accuracy [21, 22], the cross-entropy loss function was selected for its simplicity and computational efficiency in this classification-focused context [23, 24]. Therefore, a hybrid loss function was not utilized in this work.

Evaluation Metrics

The main evaluation metric observed in this study is accuracy. To comment on the accuracy (1) more precisely, the F1 score (2) is also taken into consideration as well as precision (3), recall (4), and AUC (5). The key indicators that define the performance of these metrics are true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). False positive (FP) happens when a negative sample is wrongly labeled as positive, while a true positive (TP) signifies a positive sample that is correctly classified as positive. Conversely, a false negative (FN) occurs when a positive sample is incorrectly identified as negative, whereas a true negative (TN) refers to a negative sample that is accurately identified as such [25, 26].

Table 3 Different configurations of the ViT models

Model	Learning rate	Pretrained model	Optimizer
ViT	0.0001	ViT-B/16	Adam
ViT#2	0.00002	ViT-B/16	Adam
ViT#3	0.000001	ViT-B/16	Adam
ViT#4	0.00002	ViT-L/16	Adam
ViT#5	0.00002	ViT-L/16_224	Adam
ViT#6	0.00002	ViT-L/16_224	RAdam

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$F - 1Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{2}$$

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

$$AUC = \int TPR(FPR)dFPR \tag{5}$$

Accuracy represents the ratio of correctly predicted observations to the total observations. Precision is the ratio of correctly predicted positive observations to the total predicted positive observations. It is a measure of a classifier's exactness. A high precision relates to a low false positive rate. Recall is the ratio of correctly predicted positive observations to all the observations in the actual class. It is a measure of a classifier's completeness. A high recall indicates that the class is correctly recognized (a low number of false negatives). It is also known as true positive rate (TPR). The F1 score is the weighted average of precision and recall. It is a balance between precision and recall, providing a single metric that accounts for both false positives and false negatives. It is particularly useful when the class distribution is uneven [27, 28].

Results and Discussion

Results

Models are primarily evaluated by accuracy, precision, recall, and F1 score. First, an extensive study is performed on the eye disease dataset as its test images are publicly available with corresponding labels, cataract (C), glaucoma (G), diabetic retinopathy (D), and normal (N).

Table 4 summarizes the overall performance of various DL architectures for eye disease classification. According to Table 4, Vit#5 demonstrates the highest overall accuracy (98.1%), indicating it correctly classified the most images. While ViT#4 achieves a comparable accuracy (97.5%), other metrics might be more relevant depending on this multi-class classification. For instance, if minimizing false positives (healthy eyes identified as diseased) is crucial, ViT#5 remains preferable due to its superior precision (0.962). Moreover, if capturing all positive cases

Table 4 The best overall test performances of the DL architectures

Model	Precision	Recall	F1 score	Accuracy
Base-CNN	0.862	0.856	0.856	0.928*
ResNet50	0.929	0.930	0.929	0.965*
DenseNet121	0.945	0.944	0.944	0.973*
Inception-ResNetV2	0.950	0.948	0.949	0.947*
ViT	0.880	0.871	0.872	0.941*
ViT#2	0.937	0.937	0.937	0.969*
ViT#3	0.908	0.907	0.907	0.954*
ViT#4	0.950	0.950	0.950	0.975*
ViT#5	**0.962**	**0.961**	**0.962**	**0.981**
ViT#6	0.944	0.945	0.944	0.974*

*Indicates significant p values (<0.05) for pairwise comparisons between ViT#5 and the other models (Welch's t-test)

(not missing any diseased eyes) is critical, then ViT#5 with higher recall like (0.961) still remains the best model.

Table 5 displays the model performances in terms of different retinal eye disease classes. According to this table, ViT#5 performs exceptionally well in accurately classifying normal, glaucoma, and cataracts classes. The best performing model for cataract classification is ViT#5 with an accuracy of 0.983, precision of 0.960, recall of 0.977, and F1 score of 0.858. The best performing model for DR classification is ViT#2 and ViT#6 with an accuracy of 1.000, precision of 1.000, recall of 1.000, and F1 score of 1.000. The best performing model for glaucoma classification is ViT#5 with an accuracy of 0.973, precision of 0.949, recall of 0.935, and F1 score of 0.942. While Inception-ResNet-v2 achieves a very high precision for glaucoma, its recall, F1 score, and accuracy are lower compared to ViT#5. The best performing model for normal eye classification is ViT#5 with an accuracy of 0.970, precision of 0.938, recall of 0.943, and F1 score of 0.941. ViT#5 performs well-balanced for classifying normal eyes, exhibiting strong recall, F1 score, accuracy, and precision. Vit#5 is the most consistent performer across all classes, achieving high accuracy, precision, and recall for cataract, DR, and normal classifications.

Figure 4 demonstrates four confusion matrices for CNN models. Inception-ResNet-v2 stands out for consistently achieving the highest number of true positives for two out of the four classifications: normal ($n=210$) and glaucoma ($n=185$). However, DenseNet121 takes the lead for DR classification with 242 true positives and ResNet50 takes the lead for cataract classification with 208 true positives. The choice between these two models depends on the specific application's priorities. If overall disease detection is paramount, Inception-ResNet-v2 would be preferable. On the other hand, if accurate DR classification is crucial, DenseNet121 could be a better choice.

Table 5 The best class-wise test performances of the DL architectures

Model	Cataract				Diabetic retinopathy				Glaucoma				Normal			
	Precision	Recall	F1	Acc	Precision	Recall	F1	Acc	Precision	Recall	F1	Acc	Precision	Recall	F1	Acc
Base-CNN	0.937	0.831	0.881	0.938	0.990	0.990	0.990	0.995	0.801	0.718	0.757	0.890	0.719	0.886	0.794	0.890
ResNet50	0.972	0.920	0.945	0.972	0.977	0.995	0.986	0.993	0.874	0.865	0.870	0.936	0.892	0.938	0.914	0.960
DenseNet-121	0.965	0.970	0.967	0.985	0.996	1.000	0.998	0.999	0.939	0.876	0.907	0.958	0.882	0.928	0.904	0.951
Inception-ResNet-v2	**0.976**	0.958	0.967	0.983	0.985	1.000	0.993	0.996	**0.949**	0.881	0.914	0.958	0.890	0.955	0.921	0.957
ViT	0.798	0.945	0.866	0.924	0.996	0.988	0.992	0.995	0.848	0.745	0.794	0.924	0.878	0.804	0.839	0.922
ViT#2	0.963	0.933	0.948	0.973	**1.000**	**1.000**	**1.000**	**1.000**	0.884	0.884	0.884	0.945	0.900	0.930	0.915	0.956
ViT#3	0.927	0.907	0.917	0.956	1.000	0.990	0.995	0.998	0.850	0.854	0.852	0.930	0.854	0.878	0.866	0.931
ViT#4	0.973	0.968	**0.971**	0.985	0.995	0.991	0.993	0.996	0.911	0.920	0.915	0.960	0.919	0.919	0.919	0.960
ViT#5	0.960	**0.977**	0.858	**0.983**	1.000	0.991	0.995	0.998	**0.949**	**0.935**	**0.942**	**0.973**	**0.938**	**0.943**	**0.941**	**0.970**
ViT#6	0.955	0.959	0.957	0.977	**1.000**	**1.000**	**1.000**	**1.000**	0.894	0.925	0.909	0.962	0.926	0.895	0.910	0.956

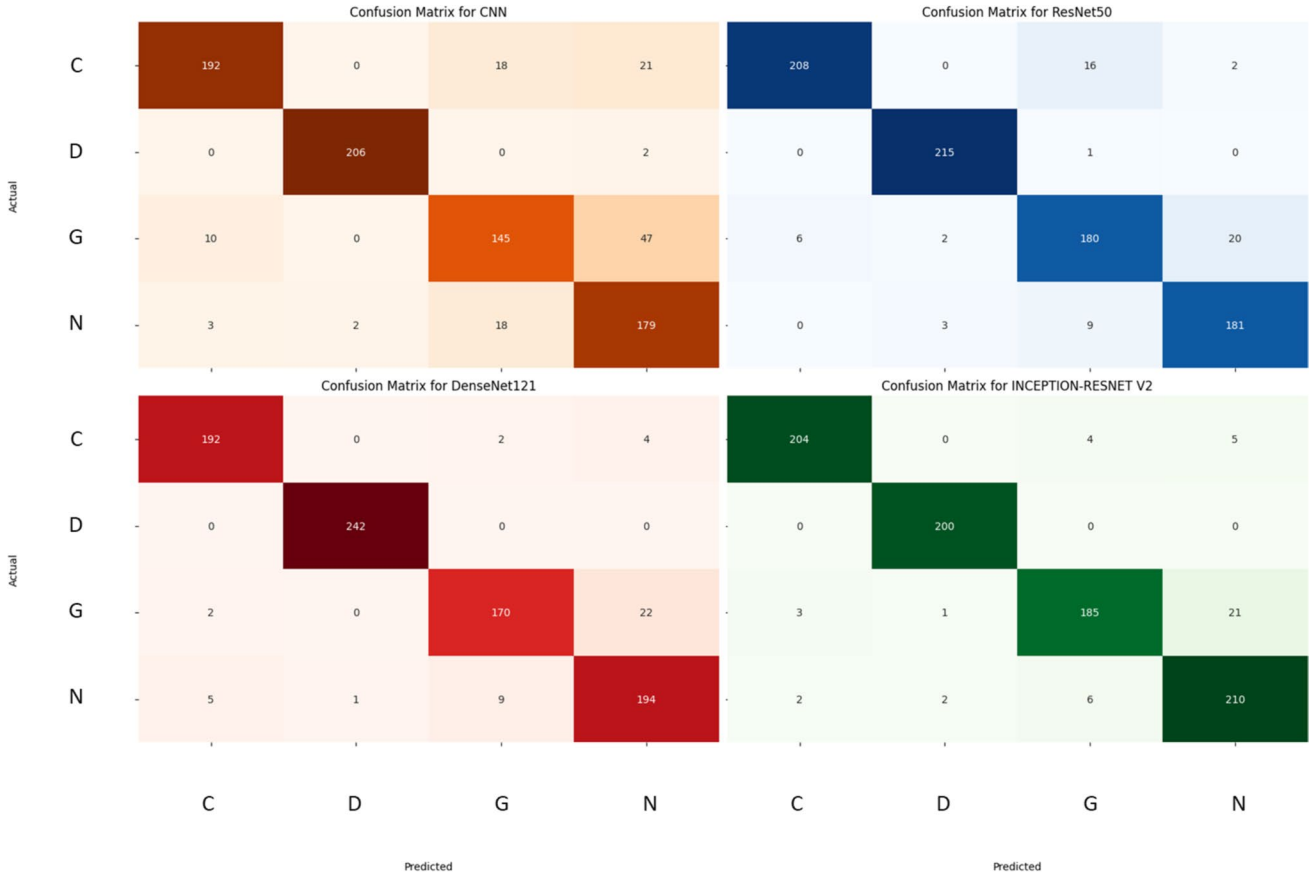


Fig. 4 Confusion matrix for CNNs

Figure 5 illustrates six confusion matrices for ViT models. Examining the true positives across all six ViT confusion matrices reveals some interesting patterns. While no single model dominates across all classifications, ViT#5 shines in cataract ($n = 217$), glaucoma ($n = 187$), and normal ($n = 198$) disease detection. Base ViT excels at DR classification ($n = 244$). Notably, ViT#3 and Vit#6 show lower true positives compared to other models. Ultimately, the choice of the best model hinges based on application's priorities. If overall disease detection is paramount, ViT#5 could be a good first choice. However, if DR is a major concern, base ViT can be chosen with the highest true positives for that particular disease.

Figure 6 shows ROC classes detected by CNNs. The ROC curves are plotted for multiple classes (C, D, G, and N), representing cataract, diabetic retinopathy, glaucoma, and normal. While Inception-ResNet-v2 appears to have the overall best performance (AUC = 0.97), it is important to consider the performance for each specific class. For instance, DenseNet121 outperforms Inception-ResNet-v2 for a cataract (AUC = 0.98) even though Inception-ResNet-v2 has a higher overall AUC for other diseases such as DR (AUC = 1.00), glaucoma (AUC = 0.93), and normal patients (AUC = 0.96).

Figure 7 shows ROC classes detected by ViTs. While ViT#4, ViT#5, and ViT#6 appear to have the overall best performance (AUC = 0.97), it is important to consider the performance for each specific class. For instance, ViT#5 outperforms other models for glaucoma (AUC = 0.96) and normal patients.

Overall, VIT#5 appears to be a strong model across different classes, especially for "cataract" and "normal," with high F1 scores, indicating a good balance between precision and recall. ViT#4 and ViT#5 have the highest accuracies in Cataract, but ViT#5 excels with the highest accuracy in the Normal category as well. ViT#2 and ViT#6 achieved perfect accuracy (1.000) in DR, which suggests very significant performance for this specific condition. DenseNet121 and Inception-ResNet-v2 perform very well across all classes, both achieving the highest overall accuracy. ResNet50 shows strong performance, particularly in the "normal" class. The initial CNN and ViT tend to have lower metrics compared to the other models, which might



Fig. 5 Confusion matrix for ViTs

indicate that the complexity and architectures of the more advanced models are better suited for this dataset.

Even though proposed model ViT#5's overall performance is significantly good, in order to address the potential issues of feature drift and catastrophic forgetting due to potential weak supervision, saliency maps for the proposed ViT#5 were generated for both success and failure cases across each class (Fig. 8). These maps illustrate the model's attention regions, helping to verify if it consistently focuses on relevant diagnostic features or if any spurious correlations have influenced the learned representations. The details on the Vit#5 model's success and failure cases (false negatives) are included in Table 6.

The saliency map analysis shows that, for successful predictions, the model consistently focuses on relevant diagnostic features, retaining valuable pretrained representations. However, failure cases reveal dispersed or irrelevant attention, indicating potential issues with feature drift and spurious correlations in weakly supervised learning. To mitigate these effects, future work could incorporate

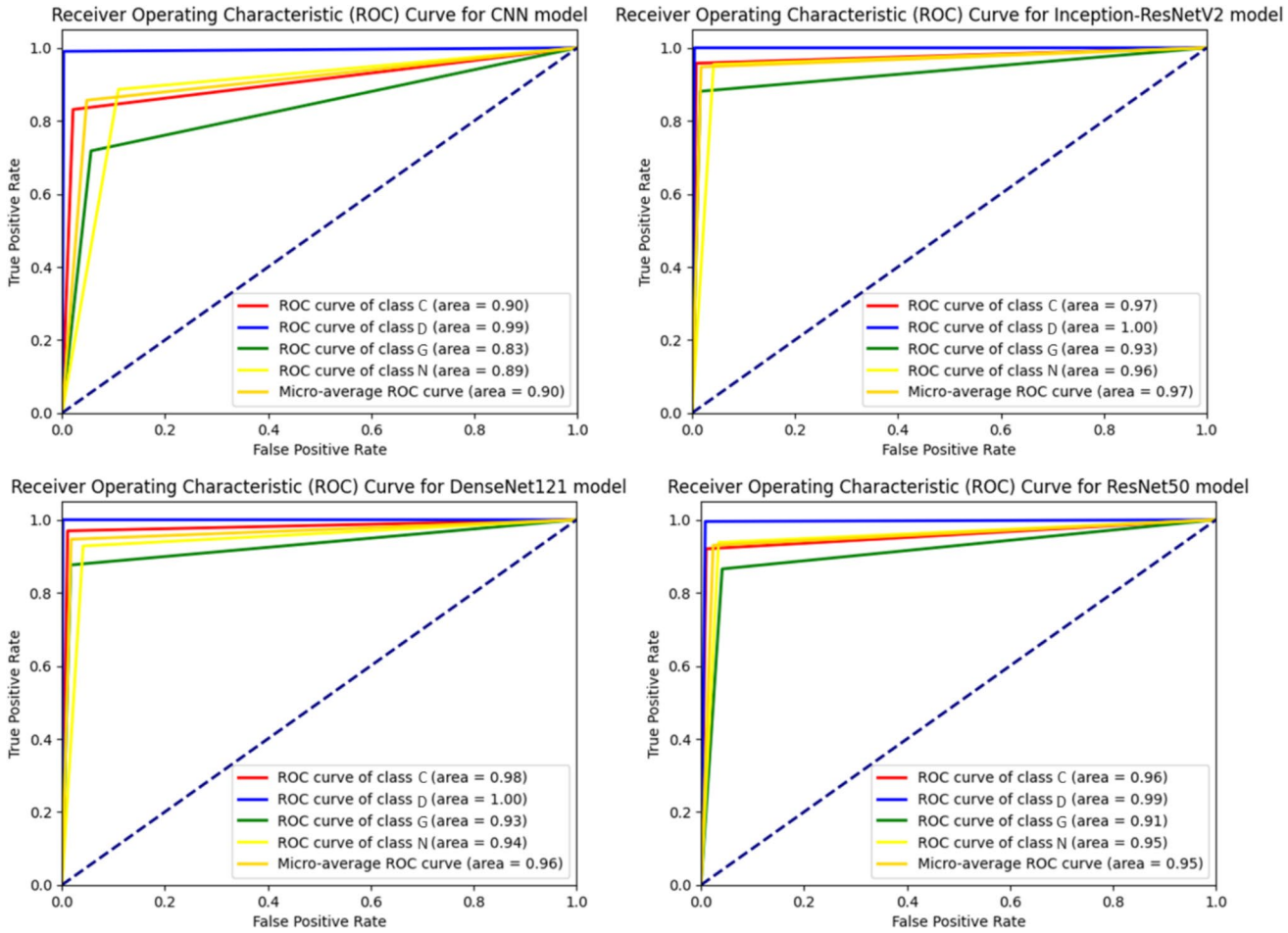


Fig. 6 Receiving operator curve (ROC) classes detected by CNNs

dynamic weight freezing, elastic consolidation, or contrastive loss to reinforce critical feature retention and prevent catastrophic forgetting. This approach would ensure that the model remains focused on clinically significant features throughout the classification process.

To validate the generalizability of our proposed model (ViT#5), we reserved 20% of the dataset as an independent test set that was completely excluded during training and validation. This portion ensured unbiased evaluation of Vit#5 model's real-world applicability. The results showed an overall accuracy, precision, and recall of 94.21%. Class-specific metrics are shown in Table 7.

These results confirm the model's strong generalizability and robustness, particularly for conditions like diabetic retinopathy, where early and accurate diagnosis is critical. However, slightly lower recall for "normal" cases suggests potential areas for refinement. This evaluation supports the clinical applicability of the proposed approach and highlights its potential for deployment in real-world scenarios.

Discussion

Several state-of-the-art classification techniques, including CNN-based algorithms, are compared to the proposed method for retinal eye disease classification to confirm its superiority (Table 8). These techniques have been tested on several datasets such as Kaggle, ODIR, RFMiD, Mendeley, and local hospital datasets. The proposed ViT#5 framework held the best accuracy among all the models tested, at 98.1%. This shows how robust the model is in comparison to other state-of-the-art techniques. The proposed framework outperforms other state-of-the-art models, including CNN [29], Two I/P VGG16 [8], EyeDeep-Net [30], EfficientNet [3], and GoogleNet [13]. With an accuracy of over 80%, the other models performed substantially better than the EyeDeep-Net, which achieved 76.1%. The experimental results demonstrate sufficient evidence of the performance enhancement obtained by

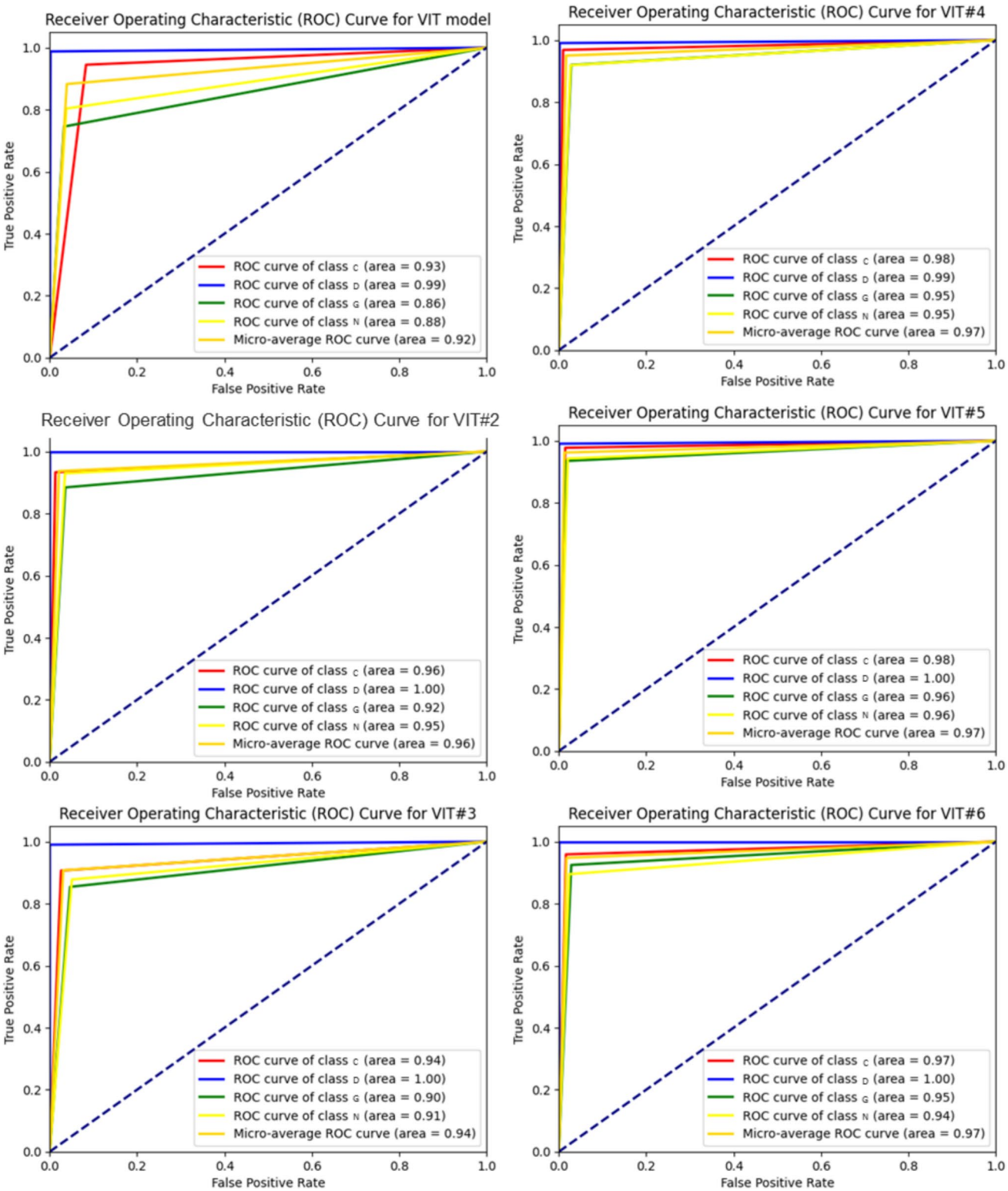


Fig. 7 Receiving operator curve (ROC) classes detected by ViTs

incorporating a transfer learning-based ViT network, while also offering insight into the limited utility of convolutional-based neural networks in robustly detecting various types of retinal eye diseases. This analysis demonstrates the superiority of the proposed framework that outlines ViT#5's potential for classifying different types of retinal

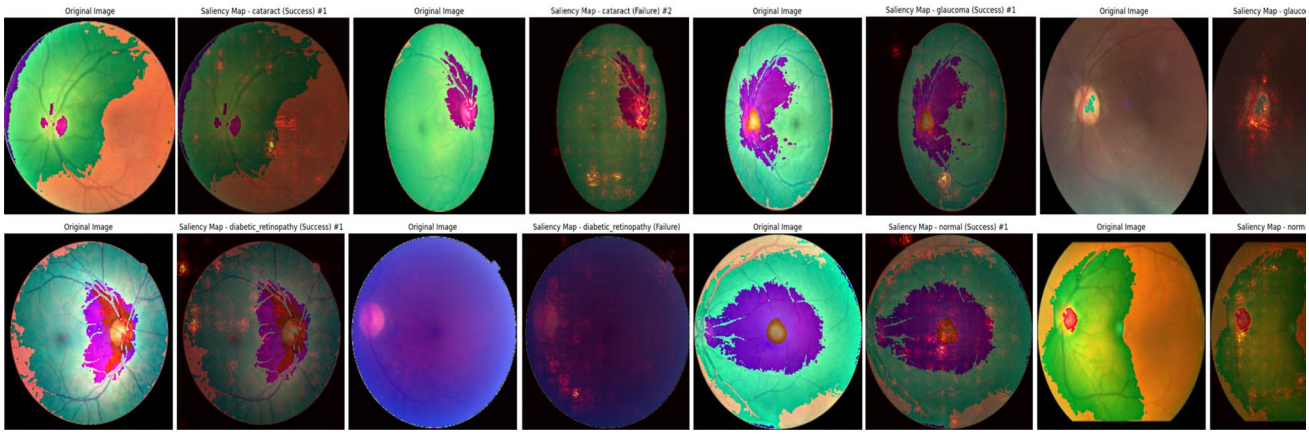


Fig. 8 Saliency maps for each class showing success and failure cases

eye disease. Table 8 also shows that, at 98.1%, our proposed framework had the highest accuracy, whereas EfficientNet and Conv-Vit achieved second and third place with an accuracy of 94.8% and 94.5% [11]. This evaluation exhibits that, regardless of the dataset being utilized, the proposed framework can yield reliable and remarkable findings.

While the vision transformer models used in this study demonstrate strong performance, they are not considered state-of-the-art (SoTA) for medical imaging tasks. Hybrid models such as Conv-ViT and CNN-trans, which integrate CNNs in the decoder, have shown superior performance for similar tasks [32, 32]. Acknowledging this limitation, future work could explore the adaptation of these models for retinal disease classification.

Recent advancements in 3D MedSAMs, such as those explored by Shen and Unberath (2024) in their 3D slicer extension for FastSAM-3D, provide accelerated labeling capabilities. However, these methods make substantial compromises, including sparse and flash attention, which do not restore native resolution. While promising for normal anatomy or specific pathologies with high target-to-volume ratios, these techniques are not yet ready for complex, multi-scale medical tasks such as those proposed in this study [33].

The slight class imbalance in the dataset (Table 2) could lead to biased classification results. To address this, future work could incorporate data augmentation techniques that enhance dataset balance and model reliability. For instance, GAN-based augmentation with hybrid loss functions, like StyleGAN and Self-Attention GAN, has been shown to effectively diversify datasets and mitigate issues such as mode collapse, ultimately improving classification accuracy and generalizability in medical imaging contexts [21, 23, 34]. The ViTs in this study utilize self-attention mechanisms but do not incorporate cross-attention, as the dataset lacks dense supervision. This patient-level annotation strategy presents a limitation, as cross-attention methods, such as Mask2Former with MP-Former modules, are optimized for densely supervised tasks and could potentially deliver improved performance [35]. Future research could address this limitation by integrating more detailed annotations.

Conclusion

In this study, ten different models from five different approaches were trained to compare the performances of various types of DL architectures for image recognition. Pretrained ViTs and state-of-the-art CNN architectures are compared for classifying retinal diseases. By leveraging transfer learning, the pretrained ViTs demonstrated superior accuracy and robustness, highlighting their potential for medical imaging tasks. This work underscores the benefits of using advanced transformer architectures over traditional CNNs for retinal disease classification. The novelty of the study lies in the implementation of transfer learning-based method with ViTs and CNNs for multi-class classification of ophthalmoscopy images without the need for high GPU capacities. As the performance metrics and visualizations show, the models performed well and closely to each other. It can be observed that using pre-trained CNN models would improve the overall performance of a base-line CNN model. It can also be observed that if the same conditions in terms of parameters and optimizers were given to the pre-trained CNN models, they would perform closely to each other. In terms of accuracy, DenseNet121 and Inception-ResNet-v2 models achieve the highest among the CNN models with a score of 97%. One of the most recent approaches for the image recognition methodology, the ViT approach also performs very well among the models in this study. The initial ViT model gives an accuracy of 94%, whereas with only a slight change of learning rate in ViT#2, the accuracy

Table 6 Observations on saliency maps for success and failure cases

Class	Success case	Failure case
Cataract	The model focuses on regions relevant to cataract features, such as central and peripheral retinal areas, showing that it has effectively retained cataract-specific features	Attention is often dispersed or directed toward irrelevant areas, suggesting difficulties in distinguishing subtle cataract features and possible confusion with other conditions
Diabetic retinopathy	Saliency maps highlight critical retinal blood vessel regions, which are important for diabetic retinopathy detection, showing effective learning of disease-specific features	The model's attention is scattered and does not focus on specific diabetic retinopathy markers. This lack of alignment suggests potential confusion and possible feature drift, where relevant features are not retained accurately in challenging cases
Glaucoma	The model consistently focuses on optic nerve regions and related areas, indicating successful retention of features critical to glaucoma identification	Failure cases show attention on irrelevant or peripheral areas, suggesting potential confusion with non-glaucoma features, possibly due to weak supervision or attention drift
Normal	In normal cases, the model's attention is appropriately spread across the retina without focusing on disease markers, indicating solid understanding of healthy retinal patterns	Failure cases occasionally highlight regions resembling disease features, leading to false positives and suggesting some overfitting to disease features in the model's learning

Table 7 Class-specific performance on unseen test data

Class	Precision	Recall	F1 score
Normal	0.92	0.89	0.91
Cataract	0.94	0.97	0.95
Glaucoma	0.90	0.91	0.91
Diabetic retinopathy	1.00	1.00	1.00

Table 8 Comparison of model performances with state-of-art methods

Model	Datasets	Results
CNN [29]	Kaggle (4 classes)	Acc: 0.813
Two I/P VGG16 [8]	ODIR (8 classes)	Acc: 0.872
EyeDeep-Net [30]	RFMiD (4 classes)	Acc: 0.761
EfficientNet [3]	2748 (2 classes)	Acc: 0.948
Conv-ViT [11]	Mendeley (4 classes)	Acc: 0.945
GoogleNet [13]	Kaggle (4 classes)	Acc: 0.927
CNN-trans [31]	Kaggle (4 classes)	Acc: 0.947
Conv-VIT [32]	Mendeley (4 classes)	Acc: 0.945
Proposed with ViT#5	Kaggle (4 classes)	**Acc: 0.981**

immediately goes up to 97%. With the learning rate modification in ViT#3, it is observed that the accuracy would go down to 95%. By changing the used pre-trained ViT model to a more in-depth model in ViT#4, the accuracy would give a higher accuracy of 97.4%. After trying the augmented-regularized pre-trained model, ViT#5 had achieved the highest accuracy among the models with 98.1%. With the same parameters and pretrained model used in VIT#5, another optimizer was also tested to observe the importance of the choice for the optimizer. As a result, this time the model ViT#6 had a lower accuracy of 97.4%. This study aimed to use a different perspective than the previous works and provide a wide usage of different approaches to collect their results under one roof. The results seem to be very close to works with other approaches and the models have a potential to improve their performances even higher. This study aimed to shed light on the importance and benefits of using AI in the medical field and how the different approaches perform in image recognition. Even though the models performed very well in this study, since the accuracy of the models are very crucial in the biomedical field, there is certainly room for improvement for future works.

Inference times pose a limitation in practical applications, particularly for resource-intensive ViT models. While CNNs can provide faster results on standard GPUs, ViTs require advanced hardware and still exhibit higher latency. Future work could consider lightweight transformers, such as those used in SAM 3D slicer modules, which have been shown to accelerate inference dramatically, even on single A6000

GPUs, facilitating near-instantaneous human-in-the-loop editing.

Due to computational constraints, the grid search and hyperparameter tuning were restricted. In the future, a more thorough grid search methodology with the more comprehensive hyperparameter tuning (i.e., with more ranges, tuning for optimizers, and learning rate) or cross-validation methods can be applied. If the cost validation and not enough computational resources would not be a constraint, different ViT pre-trained models can be used to improve the performance of retinal eye disease classification. Future extensions of this work could apply similar model adaptations to other types of medical images, specifically for organ segmentation tasks involving the liver and kidneys. Although various methods such as atlas-based approaches have been successfully applied for liver segmentation, they often face limitations due to anatomical complexity [36]. Level set-based methods, such as those demonstrated by Göçeri [19, 37], and Gaussian mixture model (GMM)-based approaches [38] have also shown promise but can struggle with variable imaging conditions and irregular organ shapes. Employing vision transformers (ViTs) and CNNs could offer more robust solutions by capturing finer details and complex spatial patterns, which are crucial for accurate segmentation of organs with irregular shapes and similar intensities to neighboring tissues. These advanced models may address challenges encountered in traditional methods [39–41] and lead to more reliable and clinically valuable segmentation outcomes.

Data Availability The datasets generated during and/or analyzed during the current study are available in the reference [14].

Declarations

Competing Interests The authors declare no competing interests.

References

1. M. H. Lee, Y. H. Chin, C.H. Ng, K.R.Y. Nistala, Z.G.W. Ow, G. Sundar, ... & C.M. Khoo, Risk factors of thyroid eye disease. Endocrine Practice, 27(3), (2021), 245–253.
2. R. Puneet Kumar, M. Gupta, Optical coherence tomography image-based eye disease detection using deep convolutional neural network. Health Information Science and Systems, 10(1), (2022), 13.
3. G. Arslan, Ç.B. Erdaş, Detection of Cataract, Diabetic Retinopathy and Glaucoma Eye Diseases with Deep Learning Approach. Intelligent Methods in Engineering Sciences, 2(2), (2023), 42-47.
4. National Eye Institute [online]. Available: https://www.nei.nih.gov/ (Accessed 21 November 2023)
5. E. Korot, M.B. Gonçalves, J. Huemer, S. Beqiri, H. Khalid, M. Kelly, M., ... & P.A. Keane, Clinician-Driven AI: Code-Free Self-Training on Public Data for Diabetic Retinopathy Referral. JAMA ophthalmology, 141(11), (2023), 1029–1036.
6. X. Zeng, H. Chen, Y. Luo, W. Ye, Automated diabetic retinopathy detection based on binocular siamese-like convolutional neural network. IEEE Access 7, (2019), 30744–30753.
7. X. Pan, K. Jin, J. Cao, Z. Liu, J. Wu, K. You, ... & J. Ye, Multi-label classification of retinal lesions in diabetic retinopathy for automatic analysis of fundus fluorescein angiography based on deep learning. Graefe's Archive for Clinical and Experimental Ophthalmology, 258, (2020), 779–785.
8. N. Gour, P. Khanna, Multi-class multi-label ophthalmological disease detection using transfer learning based convolutional neural network. Biomedical signal processing and control, 66, (2021), 102329.
9. N. Badah, A. Algefes, A. AlArjani, & R. Mokni, Automatic eye disease detection using machine learning and deep learning models. In Pervasive Computing and Social Networking: Proceedings of ICPCSN 2022 (pp. 773–787). Singapore: Springer Nature Singapore.
10. K. Karthik, M. Mahadevappa, Convolution neural networks for optical coherence tomography (OCT) image classification. Biomedical Signal Processing and Control, 79, (2023), 104176.
11. P. Dutta, K.A.Sathi, M.A. Hossain, M.A.A. Dewan, Conv-ViT: a convolution and vision transformer-based hybrid feature extraction method for retinal disease detection. Journal of Imaging, 9(7), (2023), 140.
12. T.D. Nguyen, D.T. Le, J. Bum, S. Kim, S. J. Song, H. Choo, Self-FI: Self-Supervised Learning for Disease Diagnosis in Fundus Images. Bioengineering, 10(9), (2023), 1089.
13. S. Benbakreti, U. Ozkaya,The classification of eye diseases from fundus images based on CNN and pretrained models, (2024).
14. Kaggle eye diseases dataset [online]. Available: https://www.kaggle.com/datasets/gunavenkatdoddi/eye-diseases-classification/data (Accessed: 12 August 2023).
15. K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition, (2016), 770–778.
16. G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks. In Proceedings of the IEEE conference on computer vision and pattern recognition, (2017), 4700–4708.
17. C. Szegedy, S. Ioffe, V. Vanhoucke, A. Alemi, Inception-v4, inception-resnet and the impact of residual connections on learning. In Proceedings of the AAAI conference on artificial intelligence, 31 (1), (2017).
18. Y. Bhatia, A. Bajpayee, D. Raghuvanshi, H. Mittal, Image captioning using Google's inception-resnet-v2 and recurrent neural network. In 2019 Twelfth International Conference on Contemporary Computing (IC3), 1–6, IEEE, (2019, August).
19. Göçeri, E. A Comparative Evaluation for Liver Segmentation from SPIR Images and a Novel Level Set Method Using Signed Pressure Force Function, PhD Thesis, İzmir Institute of Technology, 2013.
20. A. Dosovitskiy, L. Beyer, A: Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner,, ... & N. Houlsby, An image is worth 16x16 words: Transformers for image recognition at scale, arXiv preprint arXiv:2010, (2020), 11929.
21. Göçeri, E. Polyp Segmentation Using a Hybrid Vision Transformer and a Hybrid Loss Function, Journal of Imaging Informatics in Medicine, 2024. https://doi.org/10.1007/s10278-023-00954-2.
22. Göçeri, E. Nuclei Segmentation Using Attention Aware and Adversarial Networks, Neurocomputing, 2024. https://doi.org/10.1016/j.neucom.2024.127445.
23. Göçeri, E. An Application for Automated Diagnosis of Facial Dermatological Diseases, İzmir Kâtip Çelebi University Health Sciences Journal, 2021.

24. Göçeri, E. Vision Transformer-Based Classification of Gliomas from Histopathological Images, Expert Systems with Applications, 2024. https://doi.org/10.1016/j.eswa.2023.122672.
25. N.G. Inan, O. Kocadağlı, D. Yıldırım, I. Meşe, O. Kovan, Multi-class classification of thyroid nodules from automatic segmented ultrasound images: Hybrid ResNet based UNet convolutional neural network approach. Computer Methods and Programs in Biomedicine, 243, (2024), 107921.
26. I. Mese, N.G. Inan, O. Kocadagli, A. Salmaslioglu, D.Yildirim, ChatGPT-assisted deep learning model for thyroid nodule analysis: beyond artifical intelligence. Medical Ultrasonography, 25(4), (2023), 375-383.
27. T. Fawcett, An introduction to ROC analysis. Pattern recognition letters, 27(8), (2006), 861-874.
28. M. Sokolova, G. Lapalme, A systematic analysis of performance measures for classification tasks. Information processing & management, 45(4), (2009), 427-437.
29. R. Sarki, K. Ahmed, H. Wang, Y. Zhang, K. Wang, Convolutional neural network for multi-class classification of diabetic eye disease. EAI Endorsed Transactions on Scalable Information Systems, 9(4), (2021).
30. N. Sengar, R.C. Joshi, M.K. Dutta, R. Burget, EyeDeep-Net: A multi-class diagnosis of retinal diseases using deep neural network. Neural Computing and Applications, 35(14), (2023), 10551-10571.
31. Liu, S., Wang, W., Deng, L., & Xu, H. (2024). Cnn-trans model: A parallel dual-branch network for fundus image classification. Biomedical Signal Processing and Control, 96, 106621.
32. Mannanuddin, K., Vimal, V. R., Srinivas, A., Uma Mageswari, S. D., Mahendran, G., Ramya, J., ... & Vidhya, R. G. (2023). Enhancing medical image analysis: A fusion of fully connected neural network classifier with CNN-VIT for improved retinal disease detection. Journal of Intelligent & Fuzzy Systems, (Preprint), 1–16.
33. Shen, Y., Shao, X., Inigo Romillo, B., Dreizin, D., Unberath, M. FastSAM-3DSlicer: A 3D-Slicer Extension for 3D Volumetric Segment Anything Model with Uncertainty Quantification, arXiv preprint arXiv:2407.12658, 2024. https://arxiv.org/abs/2407.12658.
34. Göçeri, E. GAN-Based Augmentation Using a Hybrid Loss Function for Dermoscopy Images, Artificial Intelligence Review, 2024. https://doi.org/10.1007/s10462-024-10897-x.
35. Zhang, H., Li, F., Xu, H., Huang, S., Liu, S., Ni, L. M., & Zhang, L. (2023). Mp-former: Mask-piloted transformer for image segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 18074–18083).
36. Göçeri, E., Ünlü, M.Z., Dicle, O. A Comparative Performance Evaluation of Various Approaches for Liver Segmentation from SPIR Images, Turkish Journal of Electrical Engineering and Computer Sciences, 2015. https://doi.org/10.3906/elk-1304-36.
37. Göçeri, E., Ünlü, M.Z., Güzelis, C., Dicle, O. An Automatic Level Set Based Liver Segmentation from MRI Data Sets, in 2012 IEEE International Conference on Image Processing Theory, Tools and Applications, 2012. https://doi.org/10.1109/IPTA.2012.6469551.
38. Göçeri, E. Automatic Kidney Segmentation Using Gaussian Mixture Model on MRI Sequences, in Lecture Notes in Electrical Engineering, 2011. https://doi.org/10.1007/978-3-642-21747-0_4.
39. Göçeri, N., Göçeri, E. A Neural Network Based Kidney Segmentation from MR Images, in 2015 IEEE International Conference on Machine Learning and Applications, 2015. https://doi.org/10.1109/ICMLA.2015.229.
40. Dura, E., Domingo, J., Göçeri, E., Martí-Bonmatí, L. A Method for Liver Segmentation in Perfusion MR Images Using Probabilistic Atlases and Viscous Reconstruction, Pattern Analysis and Applications, 2018. https://doi.org/10.1007/s10044-017-0666-z.
41. Domingo, J., Dura, E., Göçeri, E. Iteratively Learning a Liver Segmentation Using Probabilistic Atlases: Preliminary Results, in 2016 IEEE International Conference on Machine Learning and Applications, 2016. https://doi.org/10.1109/ICMLA.2016.0104.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.